

Quiz 03

EN4660 Advanced Electronic Control Systems

Question

1. Consider the following state space representation of a system:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u, \quad y = \mathbf{C}\mathbf{x}$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = [1 \quad 0]$$

You are required to design a state feedback controller with integral action to eliminate steady-state error for a step reference input. Augment the system similar to a regulator and write down the new state-space matrices ($\mathbf{A}_a, \mathbf{B}_a$) for the augmented system. [5 marks]

2. Design a state-feedback gain $\mathbf{K}_a = [k_1 \ k_2 \ k_3]$ for the above augmented system such that the closed-loop poles are placed at $\{-2, -3, -4\}$. Compute the gain vector $\mathbf{K}_a$ using pole placement. [5 marks]
 3. Now, assume that only the output y is measurable, and the full state feedback is not available. Design a state observer for the original (non-augmented) system. Determine the observer gain matrix $\mathbf{m}$ such that the observer poles are placed at $\{-5, -6\}$. [5 marks]
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